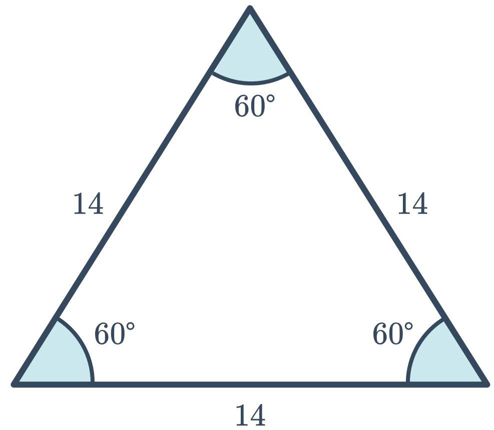
Isosceles and equilateral triangles

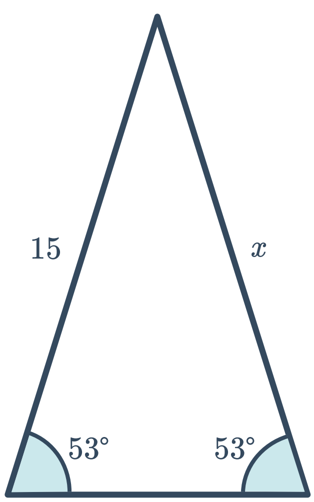
1 Consider the following triangle in the figure: Determine whether the following classifications describe the triangle.



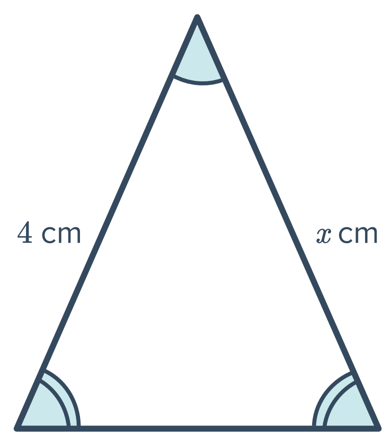
- **a** Isosceles
- **b** Scalene
- **c** Acute
- **d** Equilateral

2 Solve for x in the following triangles:

a

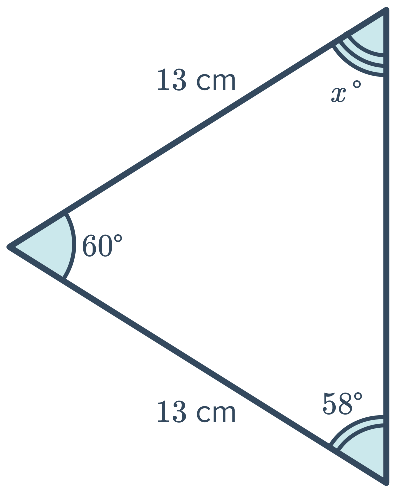


b

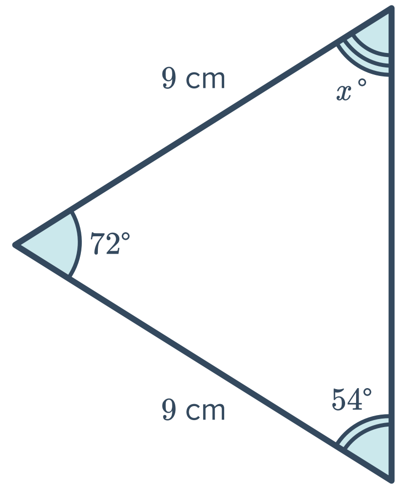


3 Solve for x in the following triangles:

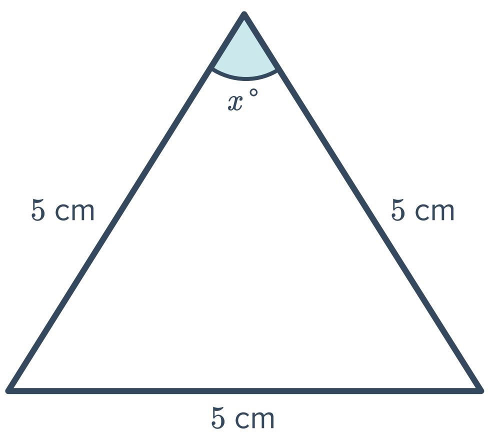
a



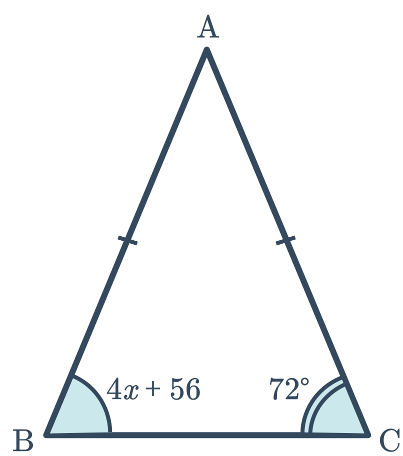
b



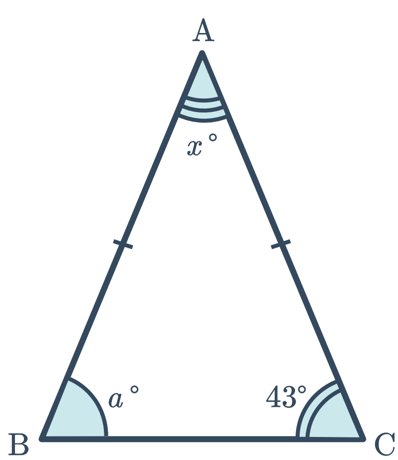
c



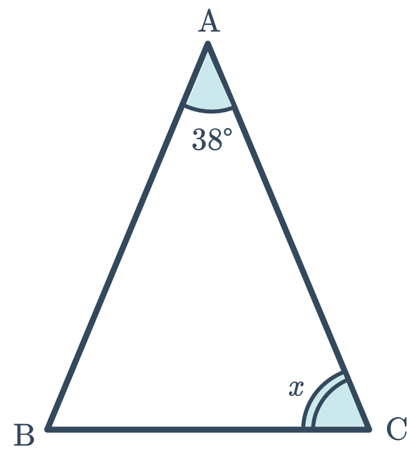
d



e



4 As shown, $\triangle ABC$ is isosceles, with $\overline{AC} \cong \overline{AB}$.
Solve for x.

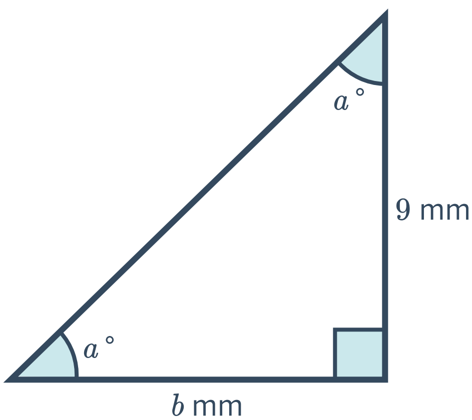


5 For each of the following triangles:

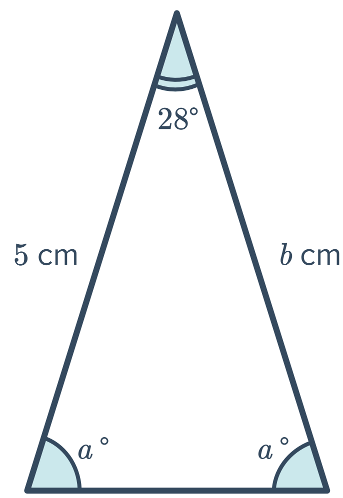
i Solve for a.

ii Solve for b.

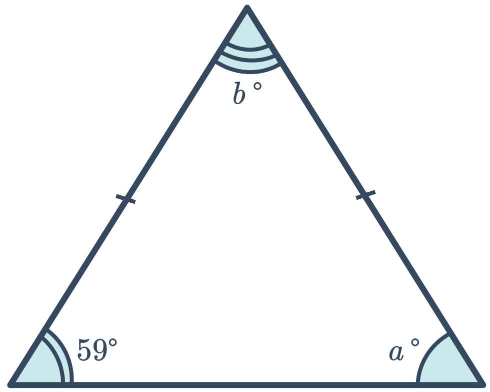
a

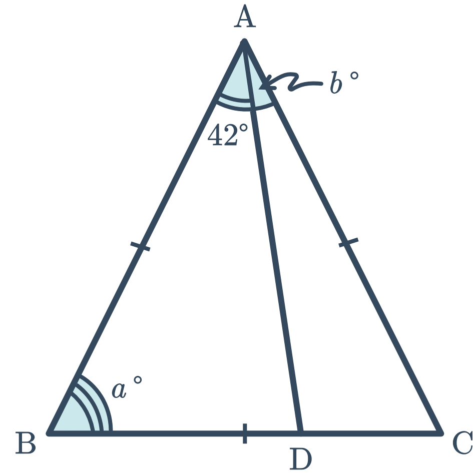


b

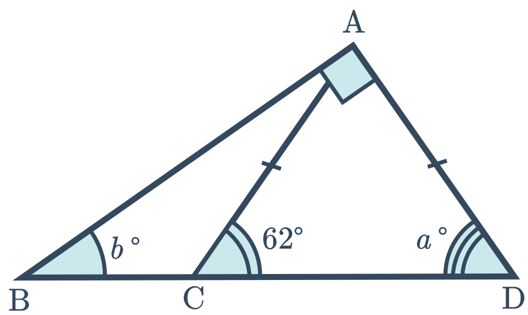


c



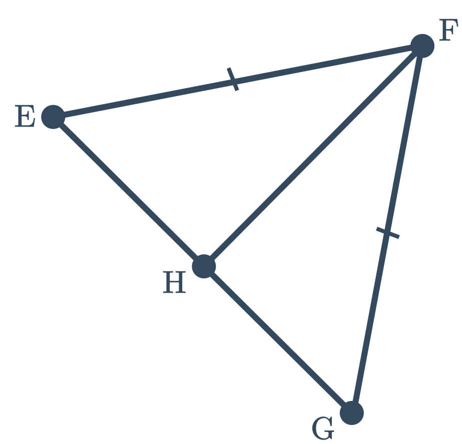


e



6 The base angles of an isosceles triangle measure y° each. The third angle measures $8y^\circ$. Solve for y.

7 Isosceles triangle $\triangle EFG$ is shown, where $\overline{FH}$ is an angle bisector.
Complete the following two-column proof to show that $\angle HEF \cong \angle HGF$:



To prove: $\angle$[redacted] $\cong \angle HGF$

	Statements	Reasons
1.	$\overline{FG} \cong \overline{EF}$	Given
2.	$\overline{FH}$ bisects $\angle EFG$	Given
3.	$\angle GFH \cong \angle EFH$	Definition of angle bisector
4.		
5.		
6.	$\angle HEF \cong \angle HGF$	Corresponding angles of congruent triangles are congruent

Additional questions

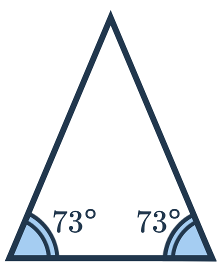
8 State the relationships between the sides and angles of an:

a Isosceles triangle

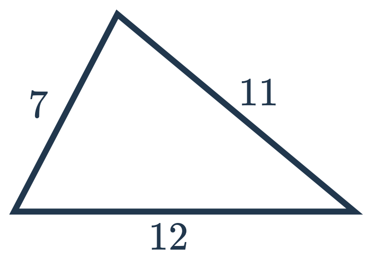
b Equilateral triangle

9 Determine if the statements are true or false for the corresponding figure:

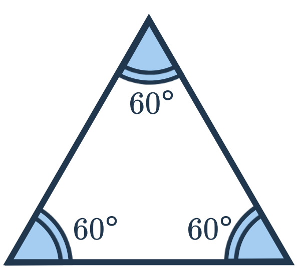
a This triangle has two congruent sides.



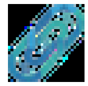
b This triangle has two congruent sides.

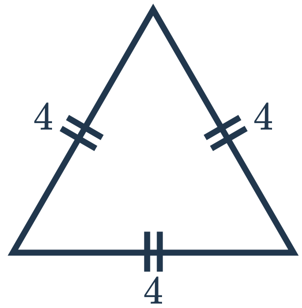


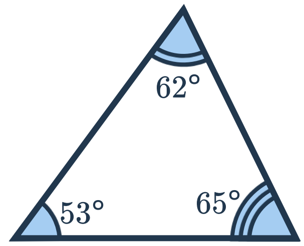
c This triangle is equilateral.



d This triangle is isosceles.

e This triangle has no congruent angles.  This triangle has no congruent sides.



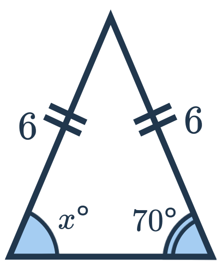


10 For each of the following triangles:

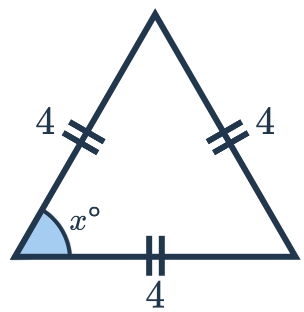
i Find the value of x

ii Justify your answer

a

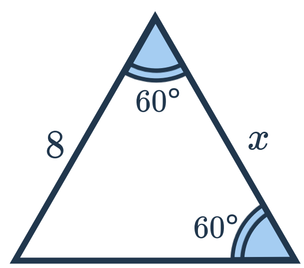


b

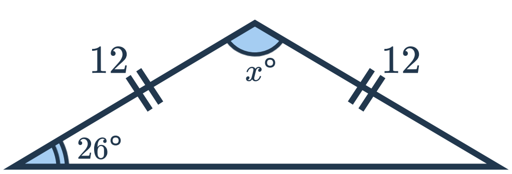


c

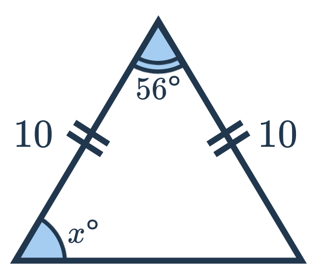
d



e



f

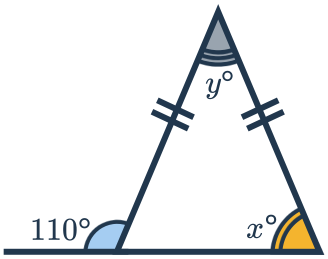


11 For each of the following diagrams:

i Find x

ii Find y

a



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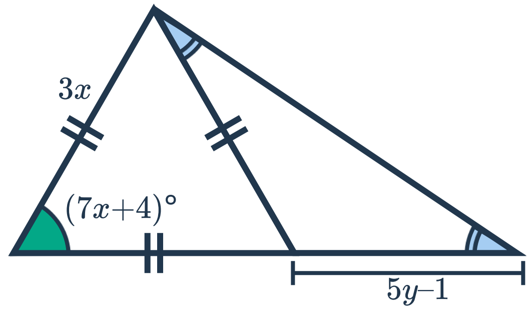


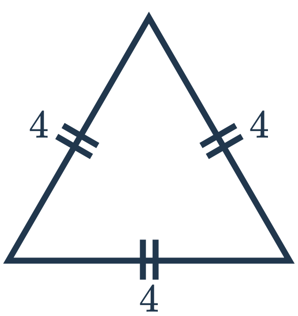
12 The lengths of the two congruent legs in an isosceles triangle are 12 inches and $(5x - 8)$ inches. Find x.

13 An isosceles triangle has a base angle with a measure of 73° and a vertex angle with a measure of y°. Find y.

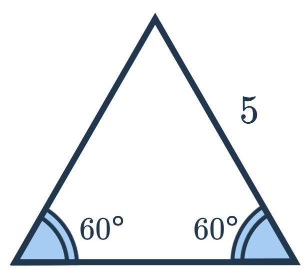
14 Two sides in an equiangular triangle have lengths of $(2x + 4)$ and $(3x - 7)$ inches. Find x.

15 Out of the following eight triangles, identify all possible pairs of congruent triangles.

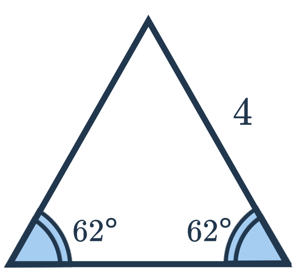
i



ii



iii



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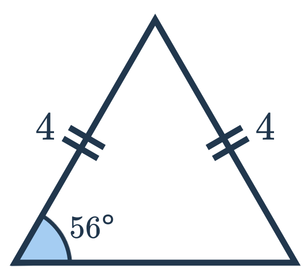

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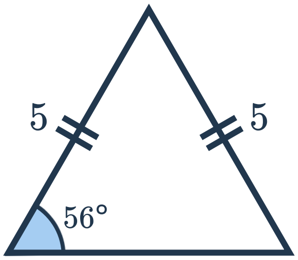
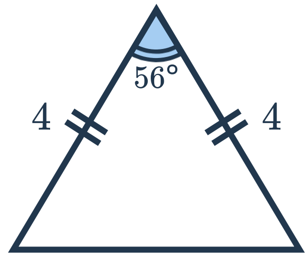
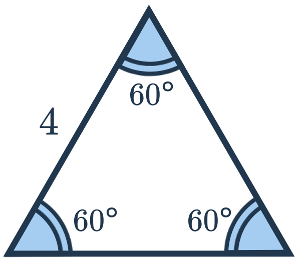
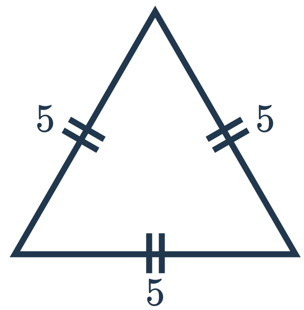




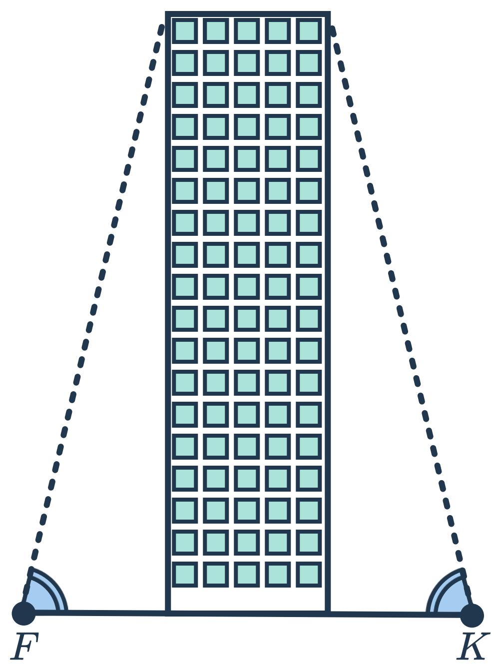
vii



viii



16 Farah and Kate are standing outside the Icon Brickell complex such that their angles of elevation to the top of the building are equal, indicated by the congruent angle markings in the diagram.
It takes Farah 20 strides to reach the base of the building, while it takes Kate 18 strides to reach the base of the building. If Kate's stride length is 30 inches, find Farah's stride length.

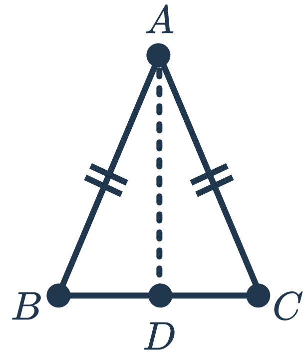


17 Determine whether the following statements are true or false. If true, justify your answer. If false, correct the statement.

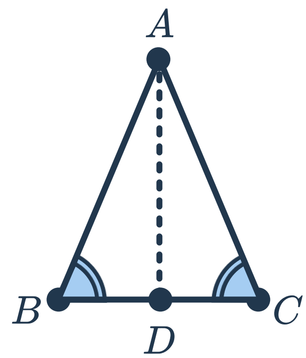
a All equilateral triangles are acute triangles.

b All isosceles triangles are acute triangles.

18 In the triangle $\triangle ABC$ we are given that $\overline{AB}$ and $\overline{AC}$ are congruent. By constructing $\overline{AD}$ to be the angle bisector of $\angle BAC$, give a proof for the base angles theorem.



19 In the triangle $\triangle ABC$ we are given that $\angle ABC$ and $\angle ACB$ are congruent. By constructing $\overline{AD}$ to be the angle bisector of $\angle BAC$, give a proof for the converse of base angles theorem.



20 Explain how the corollaries to the isosceles triangle theorems (and its converses) follow from the theorems.